

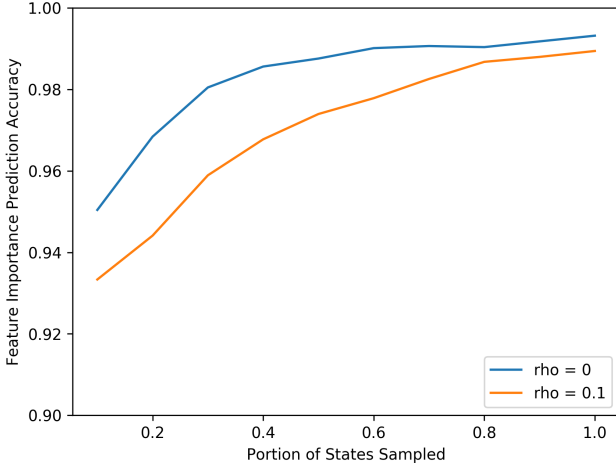


Figure 4: Comparison of feature importance prediction accuracy for increasing portion of non-terminal states.

**Domain Generation** Each domain instance is parameterized by  $\rho$  and  $m$  as specified in Section 6. For simplicity,  $d$  is always  $i_1$ . For each instance, we randomly add prerequisite relationships by selecting an item  $i_j$  uniformly at random and then an item  $i_k$  uniformly at random such that  $k > j$ . When adding prerequisite relationships, we constrain the expected number of actions to reach a terminal state to be within 10% of  $2m$ . This ensures that the domain can be solved in a reasonable amount of time using value iteration.

## 6 Experimental Results

### 6.1 Local Explanation Generalization

Based on the way we construct our abstract states, we can create “if and only if” conditions for a state in the transition sample set to be part of an abstract state’s set, as described in Section 4.3. From this, we can create a local explanation consisting of the set of features which are important in that state. To evaluate how well APG Gen can generalize when predicting important features, we generate APGs using a set of transitions less than the full set of non-terminal states (i.e., we provide a set of transitions where no  $(s, a, s')$  tuple shares an  $s$  such that only a portion of non-terminal states appear as  $s$ ). We then evaluate the local explanations by comparing to a ground truth computed for individual Pre-reqWorld instances with a domain parameter of  $m = 15$  ( $|S| = 2^{15}$ ).

The portions of correct feature classifications (important vs. not important) are shown in Figure 4. APG Gen almost always correctly identifies the important features. Even with only 10% of the states, the prediction is correct over 93% of the time. When given 80% of the states, predictions are correct 98.7% of the time for both the stochastic and deterministic environments, which suggests that the model is able to identify genuine patterns in the policy. We believe the errors the system makes are caused by the splitting order induced by APG Gen’s greedy splitting strategy.

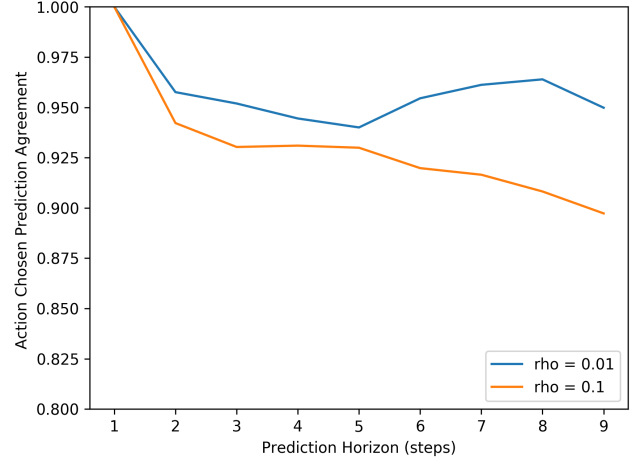


Figure 5: Action prediction for increasing time horizon.

### 6.2 $n$ -hop Prediction Evaluation

An APG is able to predict the actions an agent will take, but this ability comes from an assumption made in Section 4.1. For each pair of abstract states, we produce a transition probability: the probability that the agent will be in the second abstract state, assuming the agent is following a transition tuple chosen at random from that first abstract state. This holds for a single action for states in the provided transition sample set, but not for arbitrary states and not when performing several of these predictions in sequence.

To evaluate the error caused by making this assumption, we have APG Gen predict the distribution of actions the agent will take  $n$  time-steps in the future. We compare it to the true computed distribution and report the portion of actions for which the true and predicted distributions agree. This is for a domain parameter of  $m = 15$  ( $|S| = 2^{15}$ ). The size of the transition sample set is half the size of the set of non-terminal states.

The action prediction is consistently correct when the domain is deterministic, so we report results for two stochastic domains in Figure 5. Even with a small  $\rho$ , the prediction is less accurate as the number of steps increases, as is expected. However, there is no dramatic decrease, suggesting that the Markovian assumption made in Section 4.2 is reasonable. The steady decline is likely due to computing transition probabilities as an average of the transition sample set.

### 6.3 Explanation Size

The purpose of APGs is to be more human-interpretable than a Markov chain made from the base MDP. Therefore, the number of nodes in an APG should be much lower than the number of grounded states in the base MDP. To test this, we construct domains with a number of states ranging from 32 to 1,073,741,824 and count the number of abstract states in the corresponding APG. As in Section 6.2, for each generated APG, the size of the transition sample set is half the size of the set of non-terminal states. The results are presented in Figure 6. Note that the x-axis is in log-scale. The explanation size grows sub-linearly in  $m$  while the state-space size grows exponentially in  $m$ . This suggests that

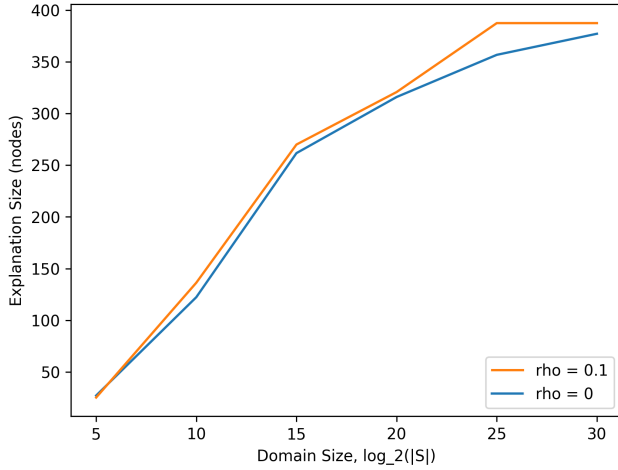


Figure 6: Comparison of explanation versus state-space size.

the explanation size is based more on the number of actions required to reach a terminal state than the number of states, which indicates that compact policy representations are being automatically extracted.

## 7 Conclusion and Future Work

We introduced Abstract Policy Graphs, a whole-policy explanation from which state-specific explanations can be extracted. In addition, we presented APG Gen, an algorithm for creating an APG given a policy, learned value function, and set of transitions, without constraints on how these are created. We showed that APG Gen runs in time quadratic in the number of features and linear in the number of transitions provided,  $O(|F|^2|tr\_samples|)$ . Additionally, we demonstrated empirical results showing the small size of the APGs relative to the original MDPs, as well as the types and quality of explanations which can be extracted. Together, these show that APG Gen can produce concise policy-level explanations in a tractable amount of time. Future work includes restructuring the explanations extracted from an APG to be better understood by a non-expert. To address this, we are in the process of conducting a user study to evaluate the usefulness of APG explanations in different presentation formats.

## 8 Acknowledgements

This material is based upon work supported by DARPA grants FA87501720152 and FA87501620042. Any opinions, findings and conclusions, or recommendations expressed in this material are those of the authors and do not necessarily reflect the views of DARPA. Hic incidunt quae dolores alias, voluptas laboriosam sint omnis eligendi qui cum, praesentium perspicatis asperiores perferendis voluptates illo mollitia recusandae ullam vero, asperiores magnam sed doloribus quod commodi natus eum rerum enim. Iure reprehenderit qui cum maiores excepturi porro consecetur, minus libero temporibus consequuntur sequi fuga numquam repellat quas excepturi voluptatibus dicta, ex itaque dicta tempora dolor non cum, voluptas dolorem molestias, placeat sapiente

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